Learning A–Z	level **N**	Multi-level	N/A
Grade	2	Word Count	543
Lexile	800L	Nonfiction • Informational	

Refer to the Focus Question on page 2 of this title to guide discussion and support additional learning connected to the text.

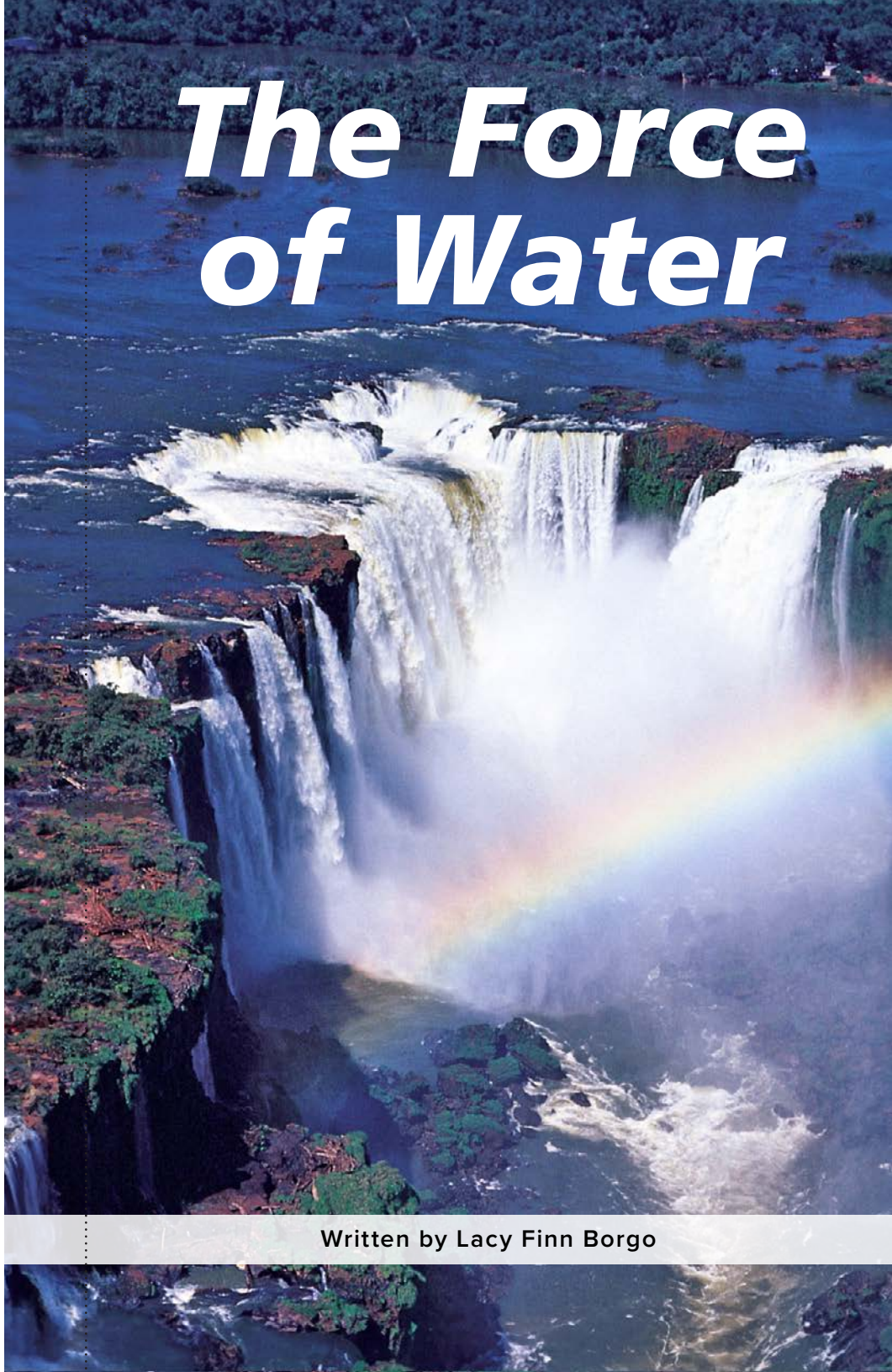
The Force of Water teaches the reader about Earth's most valuable resource—water. The book explains how water changes our planet's surface, how it changes form as it moves through the water cycle, and how and why it is important to living things. Photos and other visual aids support the text.

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The Force of Water

Written by Lacy Finn Borgo

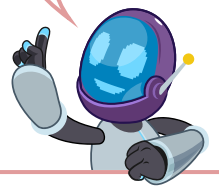
The Force of Water



Written by Lacy Finn Borgo

Focus Question

Why is it important that people take care of Earth's water?



Words to Know

deltas
floodplain
groundwater
irrigate
pollutes
sediment
tributaries
watershed
water vapor



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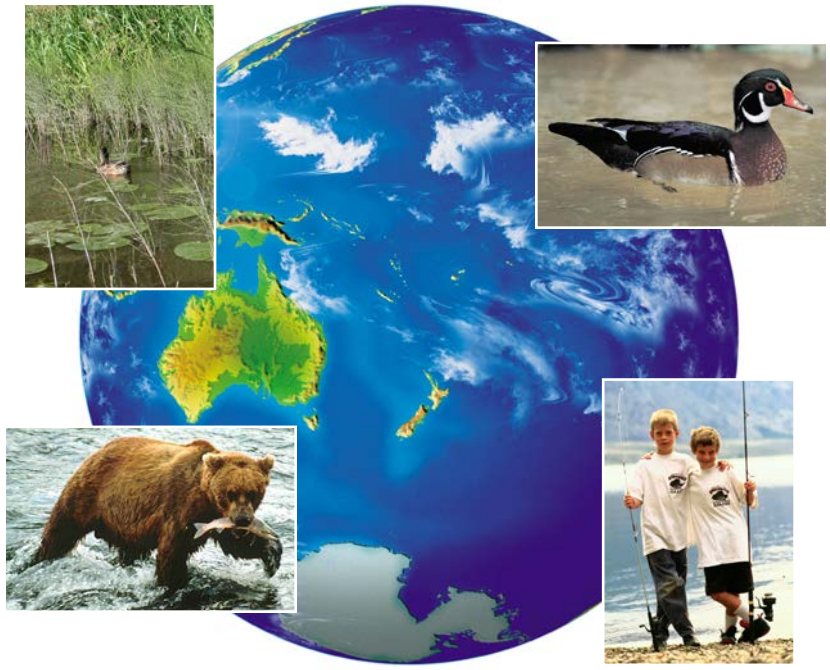
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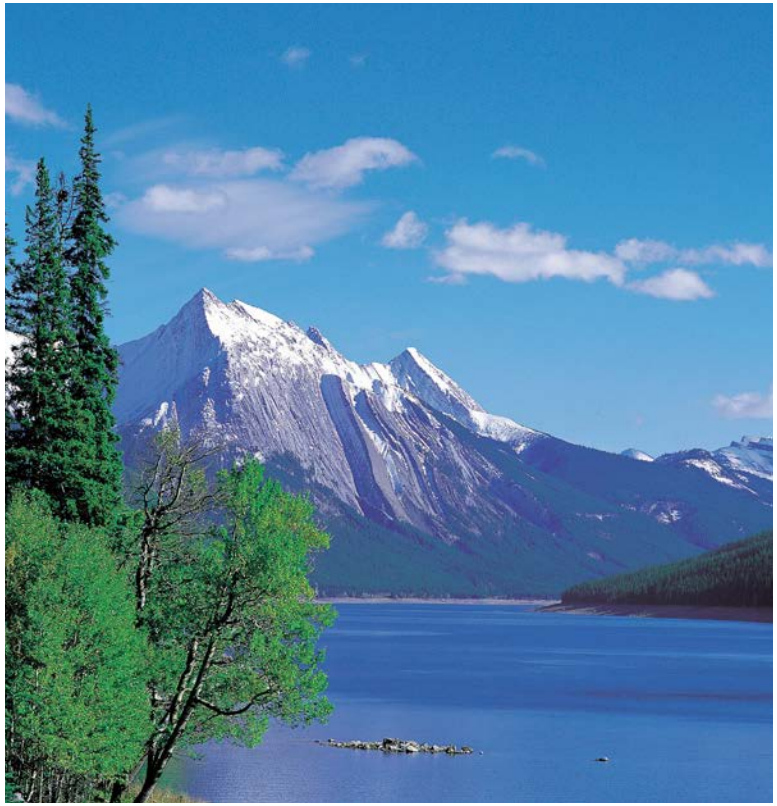
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Introduction

Water, Earth's most important resource, is a mighty force. It covers most of Earth and is always at work changing Earth's surface. Plants, animals, and humans need water to survive. So, we have to take care of the water for the good of the Earth.

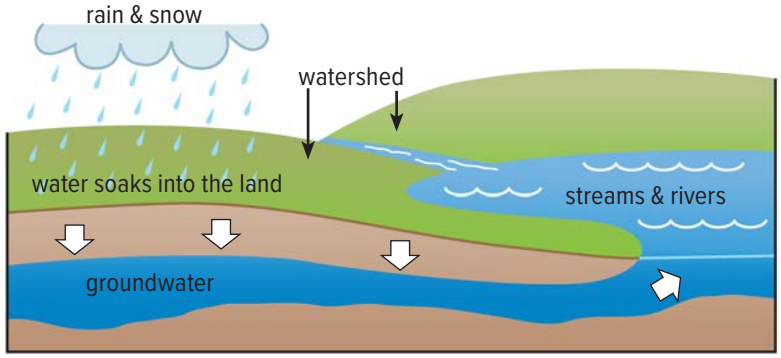
Water is found above ground in lakes, ponds, rivers, streams, and oceans. It is also found below ground, between rocks and sand. This water is called **groundwater**. The air also holds water as clouds and an invisible gas called **water vapor**.



Here you can see water in different forms. Can you name them?

Water on the Move

Much of Earth's water is on the move. Water falls from clouds as rain and snow and drains into streams and rivers. The land surrounding streams and rivers is called a **watershed**.



Streams flow into small rivers, and small rivers flow into bigger rivers. The smaller streams and rivers that flow into bigger rivers are called **tributaries**. Over time, much of the water that flows into streams and rivers reaches the oceans.



Natchez, Mississippi

The Mississippi Watershed

How many smaller rivers flow into the larger Mississippi River?

Water as a Force

As water flows in rivers and streams, it picks up dirt, sand, and rocks. The dirt and sand carried by water are called **sediment**. Sediment helps water carve away the land. The Grand Canyon is one of Earth's largest canyons. Over millions of years, it was carved by the fast-moving Colorado River. It shows how powerful water can be.



The Grand Canyon reaches a depth of more than a mile (1,600 m).



Sediment deposits form the Mississippi River Delta at the Gulf of Mexico.

When water slows down, the sediment settles to the bottom of the stream or river. Deposits of sediment form new land, called sandbars and **deltas**. Large deltas form where big rivers meet oceans.



Homes destroyed by flooding

Flooding

Sometimes massive amounts of rain and snow fall to Earth, causing streams and rivers to spill over their banks. The land around the streams and rivers floods when this happens. Flooding can cause lots of damage.

When the water recedes to the riverbed, the dirt carried by the floodwater is left on land. Over the years, this dirt forms a **floodplain**. The soil of a floodplain is rich in nutrients from the river. Crops grow well in floodplains, but can be ruined when water floods the area again.



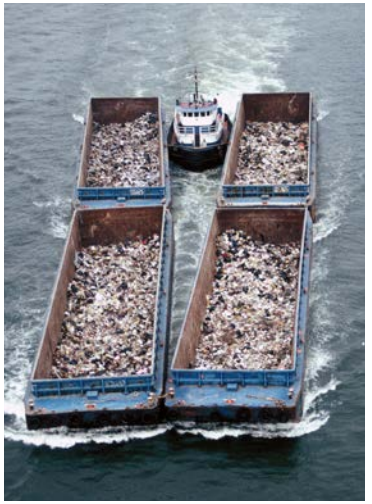
Crop land on a floodplain

Water Is Important

Rivers and streams have many uses besides carving land. People use rivers for swimming, boating, and fishing.

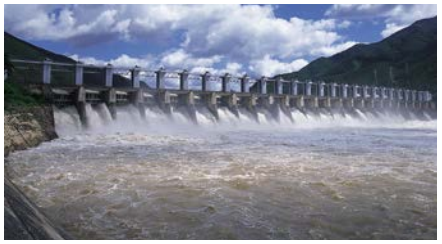


People also have used rivers for thousands of years to transport goods from place to place. Today, large barges and ships move cargo over rivers.





Lumber companies move logs from forests to mills over rivers.



Dams built on rivers use the force of water to make electricity.

Factories use millions of gallons of water each year to make the things people use.





Farms use water to **irrigate** crops.

We use water in our homes to clean and cook. How else do you use water in your home?

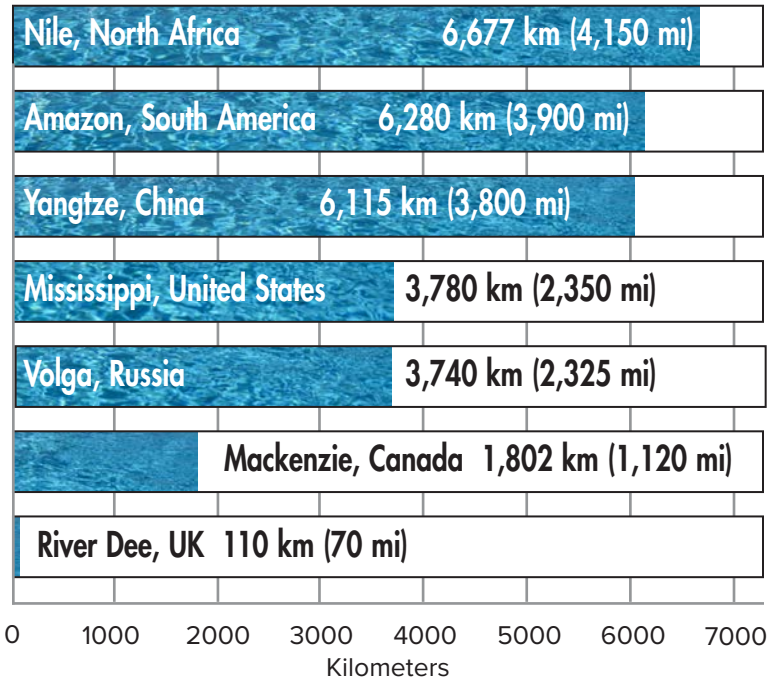


Streams and rivers are home to many living things. People and other animals depend on streams and rivers for drinking water. Some industries and large cities dump their waste into rivers, and this waste **pollutes** the water. Polluted water is harmful to humans, plants, and other animals that live in the water or use it to stay alive.



Water from factories often dumps into local waterways.

The World's Rivers



Conclusion

The force of water running in rivers and streams changes the shape of the land in many ways. Rivers and streams are important to humans, other animals, and plants. Rivers and streams will take care of us as long as we take care of them.

Glossary

deltas page 9
areas of land made by sediment at the mouths of rivers

floodplain page 11
low, flat land that floods when streams overflow

groundwater page 5
water found under the ground

irrigate page 13
to supply land with water

pollutes page 14
makes water harmful to animals, plants, and humans

sediment page 8
rocks, dirt, and sand carried by water

tributaries page 6
smaller rivers that flow into larger rivers

watershed page 6
land around a stream that drains into the stream

water vapor page 5
water in the form of an invisible gas

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